MILITARY SPECIFICATION

COURSE GYROSCOPE

(For Torpedo Mk 46 Mod 0)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of Course Gyroscope (LD 541118) for use in a torpedo. The mechanically started, electrically sustained Course Gyroscope is of one type and class, and is referred to herein as the Gyro.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

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MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-440	Soldering Techniques for Standard Type Solder Terminals
MS33586	Metals, Definition of Dissimilar

DRAWINGS

Bureau of Naval Weapons

LD 541118	Gyroscope, Course and all Documents Listed Thereon
LD 269686	Gyroscope, Course, Shipping (Packing) Condition and all Documents Listed Thereon
2047211	Gyroscope, Course

PUBLICATIONS

Bureau of Naval Weapons

QAP 60215	Quality Assurance Provisions for Gyroscope, Course

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standard for Federal Service

(Application for copies should be address to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

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3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample shall:

(a) Be manufactured using the procedures proposed for the production lot.

(b) Consist of three Gyros.

(c) Be tested as specified in section 4 herein and is for the purpose of determining that, prior to starting production, the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements.

3.2.1 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the Gyro shall be in accordance with the requirements specified herein and with the applicable documents listed on BuWeps LD 541118.

3.2.1.1 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein, and which are necessary for the execution of this specification, shall be selected in accordance with Standard MIL-STD-143, except as provided herein. AN, JAN, or MS Standard parts shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins, may be used, provided they possess suitable properties and are replaceable by the AN, JAN, or MS Standards parts without alteration, and provided the corresponding AN, JAN, or MS Standard part numbers are referenced on the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements specified herein.

3.2.2 Gages. The supplier shall provide whatever gages are necessary and adequate to assure that the material and parts of the Gyro will meet dimensions and requirements shown on the applicable documents. When requested by the supplier, the procuring activity will furnish existing pertinent drawings of Naval final inspection gages for use only in the designing of the supplier's gages. This action does not relieve the supplier of the responsibility for designing and manufacturing the gages needed for the satisfactory fulfillment of the contract requirements.

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3.2.3 Inspections and test equipment and facilities. The supplier shall furnish and maintain all necessary inspection and test equipment and facilities, and provide personnel for performing all acceptance test specified herein. The test equipment shall be adequate to permit performance of the required acceptance test for the Gyro.

3.2.4 Threads. Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28.

3.2.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used on contact with each other. Dissimilar metals are defined in Standard MS33586.

3.2.6 Safetying. Unless otherwise specified, all threaded parts shall be safetied to withstand all environmental conditions specified herein.

3.2.7 Interchangeability. All parts or subassemblies having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance.

3.2.8 Soldering. All solder connection shall be made in accordance with Standard MIL-STD-440. Where necessary, heat-sink shall be used to prevent deterioration of a component.

3.3 Performance and product characteristics.

3.3.1 Operating requirements.

3.3.1.1 Sustainer motor operating voltage. The Gyro sustainer motor voltage shall be between 73.5 ±5.7 volts peak to peak at 400 ±20 cps. Supply voltage waveform distortion of 20 percent or less shall not affect sustainer motor operation.

3.3.1.1.1 Sustainer motor surge voltage. The Gyro sustainer motor shall be capable of withstanding a surge voltage of 155 volts peak to peak for 5 seconds each time the unit is energized.

3.3.1.2 Sustainer motor current. The Gyro sustainer motor current shall not exceed 140 ma with a minimum lagging power factor of 0.9.

3.3.1.3 Yaw potentiometer operating voltage. The voltage applied to the yaw potentiometer shall be one of the following:

(a) 45 vdc, or

(b) 30 vrms at 4.8 kc ±300 cps.

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3.3.1.3.1 Yaw potentiometer surge voltage. The yaw potentiometer shall be capable of withstanding a surge voltage of 55 vrms for 5 seconds each time the unit is energized in the case of (b).

3.3.1.4 Uncaging current. Uncaging shall be accomplished by application of 1 ampere dc minimum for 15 milliseconds minimum. Uncaging circuitry before firing, shall present a 2 ± 0.7 ohm load.

3.3.1.5 Insulation resistance. The Gyro insulation resistance shall be 50 megohms minimum at 500 vdc applied for 15 seconds between isolated Gyro circuits and between these circuits and the case. This test shall not be performed more than three times on any one Gyro.

3.3.1.6 Wound and caged condition indication. The Gyro shall have a positive means of indicating wound and caged condition.

3.3.1.7 Gimbal orientation. The Gyro shall be caged with the gimbals oriented orthogonally with respect to each other within ±1 degree.

3.3.2 Performance characteristics.

3.3.2.1 Normal-plane drift rate. With the Gyro positioned in normal plane (BuWeps dwg. 2047211), the drift rate shall be limited to a maximum defined by one or the other of the following:

(a) The Gyro static drift rate shall not exceed two degrees per minute with the sustainer motor energized.

(b) A 7.5 degree Scorsby drift rate shall not exceed 2.0 degrees per minute for 9 minutes, with Gyro sustainer motor energized at 30 seconds after uncaging, and with the Gyro mounted in any direction on the Scorsby table.

(c) The yaw output excursion, during the above 9 minute Scorsby operation, shall be between 8 and 14 degrees.

3.3.2.2 Scorsby-action drift rate. The Gyro drift rate with 7.5 degree Scorsby action, when only the spring motor is operating, shall not exceed 1.5 degrees in 30 seconds after uncaging.

3.3.2.3 Yaw potentiometer wiper position. When the Gyro is caged, the yaw potentiometer wiper shall be centered on the potentiometer within one degree.

3.3.2.4 Yaw pickoff potentiometer wiper arm displacement. The yaw pickoff potentiometer wiper arm shall not be displaced more than one degree at 0.5 second after uncaging the Gyro.

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3.3.2.5 Uncaging time lapse. The time lapse from application of uncaging voltage to actual uncaging shall not exceed 100 milliseconds.

3.3.2.6 Gimbal lock. With the sustainer motor energized within ten seconds after uncaging, gimbal lock (see 6.3.8) shall not occur within 90 seconds with a 10 degree per second continuous yaw rate or 60 seconds for a 20 degree per second continuous yaw rate. The yaw rate may be in either direction.

3.3.2.7 Gimbal freedom. Gimbal freedom of both inner and outer gimbals shall be 360 degrees.

3.3.2.8 Yaw pickoff potentiometer characteristics. The yaw pickoff potentiometer shall have the following characteristics:

(a) The total dc resistance shall be 7.5 kilohms ± 5 percent.

(b) The impedance shall be 7.5 kilohms ± 6 percent at a frequency of 4800 ± 200 cps and a voltage of 26 ± 2 vrms.

(c) The active length of the potentiometer element shall extend over an arc of 358 ± 2 degrees.

(d) The linearity deviation (independent based) shall not exceed 0.5 percent of full scale.

(e) There shall be a minimum of 6 winding turns per degree of Gyro rotation.

(f) The wiper shall not bridge the end points of the potentiometer.

(g) The noise generated during operation of the potentiometer shall not exceed 200 ohms maximum, equivalent noise, at 10 microamperes.

3.3.2.9 Roll-plane excursion. A plus or minus 45 degree excursion at a rate of 4.5 degrees per second in the roll plane shall result in not more than 0.1 degree yaw excursion.

3.3.2.10 Pitch-plane excursion. A plus or minus 45 degree excursion at a rate of 4.5 degrees per second in the pitch plane shall result in not more than 3 degree yaw excursion.

3.3.2.11 Rotation-response output. The Gyro shall be capable of producing a detectable change in output in response to a rotation increment of 0.2 degree maximum in the yaw plane.

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3.3.3 Service life. The Gyro shall meet the requirement of 3.3.2.1 throughout its mission time of nine minutes.

3.3.4 Life, total operating. The unit shall be capable of satisfactory operation for a total of 25 operating cycles after receipt by the procuring activity.

3.4 Environmental requirements. Unless otherwise specified, the Gyro shall be capable of conforming to the performance and product characteristics requirements of this specification after being exposed, at any temperature in the applicable range, to the environments and under all unit conditions (see 6.3.1) specified in this section.

3.4.1 Vibration (see 6.3.2). The Gyro shall be capable of withstanding the vibration shown in table I.

Table I

Vibration Schedule

Unit Condition	Double Amplitude (in) minimum	Acceleration (g) minimum	Frequency (cps)
Non-operating	0.75	-	5-6
	-	1.3	6-16
	0.1	-	16-24
	-	3.0	24-40
	0.036	-	40-74
	-	10.0	74-500
Operating	0.050 ±0.005	-	5-20
	-	1.0 ±0.1	20-32
	0.020 ±0.002	-	32-56
	-	3.0 ±0.3	56-500

3.4.2 Shock (see 6.3.3). The Gyro shall be capable of withstanding shock specified in table II.

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Table II

Shock Schedule

Unit Condition	Peak Magnitude (g's)	Minimum Duration (ms)	Acceleration Direction
Nonoperating Ready non-operating	60	8.0	Any
Ready	45 85 300	20.0 0.5 0.5	Fwd Any Aft
Operating	85	0.5	Inertia forward

3.4.3 Acceleration (see 6.3.4). The Gyro shall be capable of withstanding the accelerations specified in table III.

Table III

Acceleration Schedule

Unit Condition	Magnitude (g)	Acceleration Direction
Nonoperating	25	Forward
Ready non-operating and Ready	35	Aft

3.4.4 Temperature (see 6.3.5). The Gyro shall be capable of conforming to the requirements of this specification at any temperature within the limits of table IV, in the unit conditions specified in table IV.

3.4.5 Pressure (see 6.3.6). The Gyro shall be capable of withstanding a pressure of 2.7 psia.

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Table IV

Temperature Schedule

Unit Condition	Temperature Range
Nonoperating	-45 to -140 degrees F
Ready, nonoperating	0 to -140 degrees F
Ready and Operating	0 to -160 degrees F

3.4.6 Humidity. The Gyro shall be capable of withstanding exposure to 95 percent relative humidity at 110 degrees F for eight hours without degradation of the finish or functional capabilities.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

3.6 Identification. The Gyro shall be permanently identified in accordance with Standard MIL-STD-130 and the applicable documents listed in BuWeps LD 541118.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. Inspections of the Gyro shall be classified as follows:

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(a) Preproduction inspection

(1) Preproduction sample (see 4.3.1)

(2) Preproduction tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Acceptance inspection

(1) Acceptance sampling (see 4.3.3)

(2) Acceptance tests and examinations (see 4.7)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of three Gyros manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the Gyro by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production tests detailed in 4.6 shall be performed at an activity designated by the procuring activity. Samples shall consist of units that have satisfactorily passed the acceptance tests and examinations detailed in 4.7; selected as follows:

(a) Step 1. Upon initiation of production, randomly select 5 units from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, proceed to step 2.

(b) Step 2. Randomly select 5 units from each group of 25 units. Upon satisfactory compliance of samples selected from three consecutive groups, proceed to step 3.

(c) Step 3. Randomly select 5 units from each group of 50 units. Continue with step 3 for the balance of the contract or until a defective unit is found.

4.3.2.1 Defective units. When a defective unit is found in any group the supplier shall:

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(a) Determine the cause of the defects.

(b) Determine if the defect is an isolated case.

(c) Take necessary action to eliminate the defects from units on hand, and if so directed by the procuring activity, from those units previously shipped.

(d) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

(e) Return to step 1 (4.3.2(a)) and proceed as indicated for inspection of units subsequently produced.

4.3.3 Acceptance sampling. The tests and examinations specified in 4.7 shall be conducted on each Gyro to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the Gyro shall be subjected to tests under the following conditions:

(a)	Temperature	Room ambient (60 to 95 degrees F)
(b)	Humidity (Relative)	Room ambient, up to 95 percent
(c)	Altitude	Normal ground atmospheric pressure

4.5 Preproduction tests. The Gyros within the preproduction sample (identified as Gyros 1, 2 and 3) shall be subjected to the tests specified in table V. Certified inspection data covering the results of preproduction tests shall be forwarded to the Bureau of Naval Weapons. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting the tests.

(b) Run or test plan used.

(c) Serial number of the Gyro.

(d) Test results of preproduction tests (4.5.2 through 4.5.3).

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform the preproduction tests specified herein:

4.5.1.1 A vibration table capable of vibrating a 9 pound load, plus fixture weight, at 10 g's maximum over a frequency range of 5 to 500 cycles per second with a maximum double amplitude of 0.75 inch.

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Table V

Preproduction Test Schedule

Test Sequence	Test Description	Computer Number		
		1	2	3
1	Acceptance Tests and Examinations (4.7).	x	x	x
2	Vibration (4.5.2.1).			x
3	Shock (4.5.2.2).	x		
4	Acceleration (4.5.2.3).			x
5	Temperature, storage (4.5.2.4.1).		x	
6	Temperature, operating (4.5.2.4.2).	x		
7	Pressure test (4.5.2.5).		x	
8	Humidity test (4.5.2.6).			x
9	Life test (4.5.2.7).	x		
10	Acceptance Tests and Examinations (4.7).		x	x

4.5.1.2 A shock testing machine capable of applying shocks of up to 300 g's maximum at durations from 0.5 to 21 milliseconds.

4.5.1.3 A centrifuge capable of exerting acceleration forces up to 35 g's magnitude for periods of three minutes minimum.

4.5.1.4 A temperature chamber capable of programmed temperature variations over a range of -45 to +140 degrees F.

4.5.1.5 A humidity chamber capable of providing a constant humidity condition of 95 percent at 100 degrees F for extended periods up to 8 hours.

4.5.1.6 Items of test equipment specified in QAP 60215.

4.5.2 Detailed preproduction tests.

4.5.2.1 Vibration

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4.5.2.1.1 Nonoperating. The Gyro shall be vibrated for a total of 9 hours (3.4.1, and tables I and VI) in accordance with the requirements of procedure XII, Specification MIL-E-5272 and the vibration deviations of 4.5.2.1.1.1.

4.5.2.1.1.1 Vibration deviations. The following deviations from procedure XII, Specification MIL-E-5272. Individual resonance surveys shall be conducted with vibration applied along each of three mutually perpendicular axes. The resonant and cycling periods in each axis shall be in accordance with table VI, with one hour at the appropriate low temperature (-45 degrees F), one hour at room temperature, and one hour at 140 degrees F.

Table VI

Resonant and Cycling Schedule

(Nonoperating)

No. of Resonances	Time at Resonance at Each Temperature (Minutes)*	Cycling Time at Each Temperature (Minutes)
0	0	60
1	9	51
2	18	42
3	27	33
4	36	24

*Nine minutes at each resonance at each temperature.

4.5.2.1.2 Operating. The Gyro shall be conditioned at room temperature for a minimum of one hour and subject to the vibration frequencies and amplitudes specified for operating condition of table I, along each of three mutually perpendicular axes. The log frequency sweep between 5 and 500 cycles shall be conducted in 7.5 minutes. The log frequency sweep between 500 and 5 cycles shall be conducted in 7.5 minutes. The log frequency sweep between 500 and 5 cycles shall be conducted in 7.5 minutes. The Gyro must be rearmed and fired prior to each sweep. The remainder of the 9-minute gyro-run time in both cases at resonance points shall be as listed in table VII.

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Table VII

Resonant and Cycling Schedule

(Operating)

Number of Resonances	Time at Resonance	Total Time
0	0	7.5
1	0.5	8.0
2	1.0	8.5
3	1.5	9.0

4.5.2.2 Shock

4.5.2.2.1 Nonoperating. Mount the Gyro onto a shock test fixture so that a minimum of 90 percent of the table input of the fixture will be applied to the Gyro under test. Subject the Gyro to the shocks listed in table VIII as specified in the Shock Test, Procedure IV, Specification MIL-E-5272 except as follows: Shock magnitude and duration shall be measured by an accelerometer mounted on the input table of the fixture to indicate applied shock. Test equipment is not restricted to that specified in Specification MIL-E-5272.

Table VIII

Shock Schedule

(Nonoperating)

Peak Magnitude (g's)	Duration (ms)	Impact Direction*	Number of Pulses
300	0.5 to 1.0	Longitudinally Aft.	3
85	0.5 to 1.0	Any direction except Longitudinally Aft.	6
60	8.0 to 8.5	Any.	9
45	20.0 to 21.0	Longitudinally Fwd.	3

*For identification of the Fwd and Aft planes refer to BuWeps Dwg 2047211.

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4.5.2.2.2 Operating. With the Gyro mounted as specified in 4.5.2.2.1, fire the Gyro and while operating, subject it to the shocks listed in table IX.

Table IX

Shock Schedule

(Operating)

Peak Magnitude (g)	Duration (ms)	Impact Direction	Number of Pulses
85	0.5	Longitudinally Fwd.	3

4.5.2.3 Acceleration. The Gyro nonoperating shall be subjected to the acceleration magnitudes and directions required by table III for one minute at room temperature for each condition specified.

4.5.2.4 Temperature.

4.5.2.4.1 Temperature, storage. The Gyro preserved and packaged in accordance with 5.2.1 shall be placed in a temperature controlled chamber and subjected to the following temperature cycle:

(a) Heat to +140 +5 -0 degrees F in 2.5 ±0.5 hours

(b) Maintain at +140 +5 -0 degrees F for 4 hours minimum

(c) Cool to -45 +0 -5 degrees F in 2.5 ±0.5 hours

(d) Maintain at -45 +0 -5 degrees F for 4 hours minimum

(e) Return to room temperature in 2.5 ±0.5 hours.

Repeat this schedule for a minimum of three cycles.

4.5.2.4.2 Temperature, operating. The Gyro shall be stabilized at zero degrees F and tested to conform to the requirements of 3.3.1 and 3.3.2. The Gyro shall then be stabilized at +160 degrees F and tested to conform to the requirements of 3.3.1 and 3.3.2.

4.5.2.5 Pressure test. The Gyro shall be subjected to a pressure of 2.7 psia. The Gyro shall meet the performance requirements specified in 3.3 after the pressure test.

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4.5.2.6 Humidity test. The Gyro shall be placed in a suitable test chamber and be subjected to 95 percent relative humidity for a period of eight hours at a temperature of 110 degrees F.

4.5.2.7 Life test. The Gyro shall be operated through 25 operating cycles (see 6.3.7). The Gyro shall be capable of meeting the performance requirements specified in 3.3 during and at the conclusion of the life test. The operating cycles accumulated during environmental tests shall be included in the required 25 cycles.

4.5.3 Preproduction acceptance criteria. Acceptance shall be based on satisfactory completion of test sequence item 10 table V.

4.6 Periodic production tests. The periodic production sample, selected in accordance with 4.3.2, shall be subjected to the tests of 4.6.2.

4.6.1 Periodic production test equipment. The following items of test equipment are required to perform the periodic production test specified herein:

(a) Items of test equipment specified in QAP 60215.

(b) Complex wave vibration machine. The vibration machine shall be capable of vibrating a weight of 40 pounds at a fundamental frequency of 50 cycles per second with a peak magnitude of 4.5 ±0.5 g. Power density of the harmonic content of the complex wave shall be reasonably distributed between 50 and 2000 cycles per second. Maximum acceleration for any 10 cps band in the specified range shall not exceed 3.5 g rms.

4.6.2 Periodic production test schedule. Subject the Gyro to the following tests in the order indicated.

(a) Vibrate the Gyro for a period of 4 to 5 minutes on the complex wave vibration machine specified in 4.6.1(b).

(b) Acceptance tests and examinations (see 4.7).

4.7 Acceptance tests and examinations. Unless otherwise specified each Gyro delivered under contract shall be subjected to the tests and examinations contained in QAP 60215. Failure to meet any requirement specified therein shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be level A or C as specified in 6.4.

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5.1.1 Level A. Cleaning, drying, preservation, and packaging methods for level A shipment shall be in accordance with the instructions contained in BuWeps LD 269686.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation and packaging shall be in accordance with the supplier's commercial practice, when such practices have been proven by testing to conform to all applicable requirements.

5.2 Packing. Packing shall be Level A or C as specified in 6.4.

5.2.1 Level A. Multiples of the Gyro preserved and packaged in accordance with 5.1.1, shall be packed in accordance with Specification MIL-P-116, Method III.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The Gyro, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Normal marking. In addition to the markings required by the contract or order, Gyro packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Description and intended use.

6.1.1 Description. The Gyro consists of four major components; the rotor and gimbal assemblies, the spring motor, the piston motor, and the yaw pickoff potentiometer. In operation, an uncage signal is applied to the piston motor which causes the piston motor to fire and release the spring motor. The spring motor overcomes the inertia of the rotor and brings the rotor up to operating speed. Series wound field coils receive an alternating voltage and sustain rotor operation. The yaw pickoff signal is provided by the yaw potentiometer. The potentiometer wiper is free to rotate 360 degrees.

6.1.2 Intended use. The Gyro is intended for use in a torpedo control system. The Gyro will provide yaw pickoff signal to the control system.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Required level of preservation and packaging.

(c) Required level of packing.

(d) Assigned activity for preproduction and periodic tests.

(e) Any exceptions to this specification.

(f) All additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions.

6.3.1 Unit condition.

6.3.1.1 Nonoperating. The condition of the uninstalled Gyro when stored, transported, or handled.

6.3.1.2 Ready, nonoperating. The condition of the Gyro when assembled with the end item, transported and handled.

6.3.1.3 Ready. The condition of the Gyro when fully assembled with the end item and installed on the launching mechanism prior to operating.

6.3.1.4 Operating. Operating is the condition of the Gyro when performing its required function.

6.3.2 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (cps).

6.3.3 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), duration in units of time (milliseconds (ms)), and a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent a maximum slope, intersect the zero g line.

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6.3.4 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g's).

6.3.5 Temperature. The temperature of the environment to which the Gyro may be exposed, expressed in degrees Fahrenheit (°F).

6.3.6 Pressure. The pressure of the environment to which the Gyro may be exposed, expressed in pounds per square inch absolute (psia).

6.3.7 Operating cycle. An operating cycle begins with the uncage signal and includes 9 minutes of operation, the removal of power, removal of the piston motor, recaging, replacement of the piston motor and reassembly and sealing of the Gyro.

6.3.8 Gimbal lock. Gimbal lock is defined as the point where the spin axis and outer gimbal axis are aligned, resulting in loss of gyroscope action.

6.3.9 Uncaged. Uncaged is the condition at which the Gyro will produce an output proportional to angular deflection.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A — This level shall be used for those items which are to be shipped to intermediate destinations or stored under indeterminate conditions for redistribution anywhere.

Level B — Not applicable.

Level C — This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-81064(Wep) COURSE GYROSCOPE (FOR TORPEDO MK 46 MOD 0)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Bureau of Naval Weapons
Washington, D. C. 20360

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